

Kleene Closure

$$\Sigma^* = \bigcup_{i=0}^{\infty} \Sigma^i = \Sigma^0 \cup \Sigma^1 \cup \Sigma^2 \cup \Sigma^3 \cup \dots$$

e.g., for $\Sigma = \{0,1\}$, Σ^* contains all binary strings

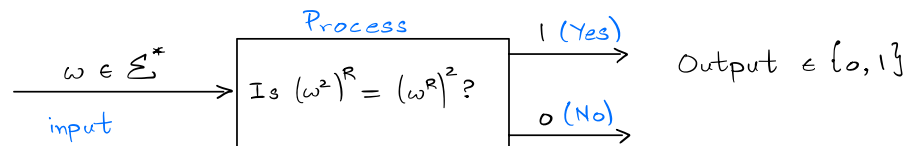
Let $w \in \Sigma^*$, where $\Sigma = \{0,1\} \rightarrow$ so, w is any possible binary string.

$w^R \rightarrow$ Reversal operation

e.g., $w = 01101 \rightarrow w^R = 10110$

For a $w \in \Sigma^*$, is $(w^2)^R$ always equal to $(w^R)^2$?

The process we use to identify which w satisfy the above criterion can be thought of as a computation.



So, we can model computation as solving a decision problem.

It is a mapping from Σ^* to $\{0,1\}$.

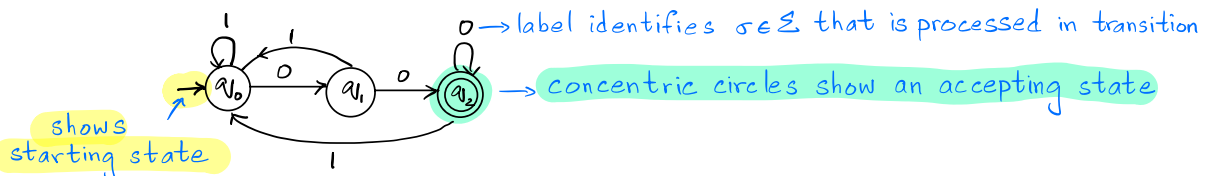
The study of computation then is to study this mapping & its properties under various restrictions.

Every computation identifies a language, i.e., the set of strings for which the output is 1.

Language Any subset of Σ^* is a language.

$$\Sigma = \{0,1\}$$

$$L_1 = \{w \in \Sigma^* \mid w \text{ ends with } 00\}$$



Deterministic Finite Automata (DFA)

A DFA is given by a 5-tuple $(Q, \Sigma, \delta, q_0, F)$

- Q is a finite set of states, e.g., $Q = \{q_0, q_1, q_2\}$

- Σ is the alphabet

- $\delta: Q \times \Sigma \longrightarrow Q$ is the transition function

e.g., $\delta(q_0, 0) = q_1$

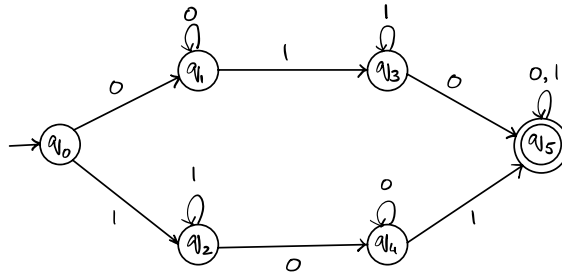
- q_0 is the starting state

- F is the set of accepting states, $F \subseteq Q$

δ	0	1
q_0	q_1	q_0
q_1	q_2	q_0
q_2	q_2	q_0

e.g.,

$L_2 = \{w \in \Sigma^* \mid w \text{ contains } 01 \text{ and } 10\}$



$L_3 = \{w \in \Sigma^* \mid w \text{ has equal occurrences of } 01 \text{ and } 10\}$

